KARUNA VASU

PYTHON DEVELOPER

CONTACT

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EDUCATION

04/2021 - 01/2023
Master of Science: Bioinformatics
Manipal Academy of Higher Education, Manipal, India
GPA: 7.24

- [Electrocardiogram Heart Rate Variability Signal Analysis and Data Acquisition] - Research Project

04/2018 - 01/2021
Bachelor of Science: Bioinformatics
Sardar Patel University, Anand, India
GPA: 7.00

01/2017 - 01/2018
XII: Science
(BSER), Rajasthan, India

01/2015 - 01/2016
X
(KSEEB), Karnataka

PROFESSIONAL SUMMARY

Results-driven Python Developer with hands-on experience in designing scalable microservices, automating complex healthcare workflows, and deploying cloud-native solutions using Django, Flask, and RESTful APIs. Proven track record in reducing processing time by up to 95% and improving system performance by up to 40% through efficient backend design, test automation, and database optimization. Adept at building robust, production-ready systems with a strong focus on performance, reliability, and clean code practices.

SKILLS

- Python Development
- Automation Engineering
- Database Management
- CI/CD Pipelines
- System Design
- RESTful APIs
- Microservices Architecture
- Cloud Computing
- Web Scraping
- Agile Methodologies

WORK HISTORY

December 2023 - Current
Python Developer, ***Integrity Healthcare Solutions Pvt Ltd***, Ahmedabad, India

- Engineered and implemented Python automation solutions at Availity, slashing claims processing time by 95%, reducing it to 45 minutes for 240 claims, and enhancing the efficiency and accuracy of claims management workflows.
- Achieved a 30% boost in operational performance by optimizing data exchange processes for major healthcare providers on the Availity platform.
- **Developed and deployed microservices architecture for claims processing, integrating with AWS-based services to improve scalability, resulting in a 40% improvement in system reliability.**
- Designed and executed comprehensive workflows for Practice Fusion's #1 cloud-based EHR platform, improving patient account creation, document management, and doctor visit scheduling.
- Increased workflow efficiency by 40% using Python, significantly

enhancing platform robustness and usability.

- Drastically cut manual testing time by 80% through the development of efficient test scripts and conducting thorough code issue resolution reviews.
- **Designed and maintained database schemas and optimized SQL queries for efficient data access and reporting across patient modules.**
- Contributed to the development of key functionalities within the eClinicalWorks platform, focusing on revenue cycle management and patient account processes.
- Utilized Python to diminish operational delays by 25%, markedly elevating the user experience and streamlining process efficiency.
- Reduced manual testing time by 60%, demonstrating a significant improvement in the development lifecycle and quality assurance processes.
- Designed and implemented robust, scalable back-end services using Python frameworks like Django or Flask.
- Developed RESTful APIs for web applications to facilitate client-server communication.
- Employed unit testing and Test-Driven Development (TDD) practices to ensure code quality and reliability.
- Optimized application performance through profiling, code optimization, and system tuning.
- **Configured and deployed Python microservices to cloud platforms such as AWS, improving deployment speed and fault tolerance.**
- Maintained and updated existing codebases to improve functionality and address bugs.

Data Science Intern, ***Zephyr Technologies and Solutions PVT LTD***, Mangalore, India

- Real-Time Chat Application
  - Developed a real-time chat application using Python and Django.
  - Implemented WebSocket connections for real-time messaging.
  - Integrated with PostgreSQL for database management.

Research Intern, ***Manipal Institute of Technology***, Manipal, India

PUBLICATION

Vasu, K., Kumar, P., Mane, P. R., Paul, B., Vaishali, K., & Sinha, M. K. (2023, September).
Electrocardiogram Heart Rate Variability Signal Analysis and Data Acquisition: A Novel
Approach for Sympathovagal Inference. In 2023 International Conference on Network,
Multimedia and Information Technology (NMITCON) (pp. 1-8). IEEE.
Presented the paper at the conference and was honored as the Best

Presenter. Received
the Best Paper Award for outstanding contribution to the field.

- Electrocardiogram Heart Rate Variability Signal Analysis and Data Acquisition.
- Database development for microfluidics.

ACCOMPLISHMENTS

- Jira Fundamentals Badge - Atlassian
- Jira Fundamentals Assessment - Atlassian
- KPMG_Data Analytics Consulting Virtual Internship - Forage
- Deep Learning - Manipal Institute of Technology
- Skills for Healthcare Data Analysts - LinkedIn
- Linux System Engineer: Advanced Disk Systems and System Backup - LinkedIn
- SAS Essential Training: Descriptive Analysis for Healthcare Research - LinkedIn
- The Non-Technical Skills of Effective Data Scientists - LinkedIn
- Tableau Essential Training - LinkedIn
- Machine Learning for Intelligent Systems - IEEE Robotics and Automation Society
- Google Foundations: Data, Data, everywhere - Coursera
- Drug Discovery - Coursera
- JavaScript Basics - Coursera
- Clinical Trials Data Management and Quality Assurance - Coursera

PUBLICATIONS

Electrocardiogram Heart Rate Variability Signal Analysis and Data Acquisition: A Novel Approach for Sympathovagal Inference, Vasu, K., Kumar, P., Mane, P. R., Paul, B., Vaishali, K., Sinha, M. K., 2023 International Conference on Network, Multimedia and Information Technology (NMITCON), 09/01/23, Presented the paper at the conference and was honored as the Best Presenter. Received the Best Paper Award for outstanding contribution to the field.

BIOINFORMATICS SKILLS

- BLAST
- NGS data analysis
- Docking